

Khwaja Yunus Ali University

School of Science and Engineering
Department of Computer Science and Engineering

Mid-term Examination: Spring 2022

Program: B.Sc. in CSE (Batch – 13th) 1st Year 1st Semester

Course Title: Physics - I

Course Code: PHY 1101

Time: 10 Minutes

Total Marks: 10 X 1 = 10

Student ID:	Batch:	Session:
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Part 1: MCQ

These are single best answer type of questions. Each of the following question or statement has five suggested answers or completions. Write down the most appropriate one on your answer script.

- Which is the unit of magnetic flux?
 - Volt*
 - Am^{-1}
 - Am^{-2}
 - JC^{-1}
 - Wb*
- The negative gradient of electric potential is equal to
 - voltage.
 - current.
 - potential energy.
 - electric force.
 - electric field.
- $\int \vec{J} \cdot \overrightarrow{da} = ?$
 - V*
 - I*
 - E*
 - R*
 - C*
- Which of the following can be a vector quantity?
 - Time
 - Area
 - Charge
 - Permittivity
 - Current
- If the surface does not enclose the charge, then the $\oint_s \vec{E} \cdot \overrightarrow{da} = ?$
 - 0
 - $\frac{\epsilon_0}{q}$
 - $\frac{q}{\epsilon_0}$
 - $\epsilon_0 q$
 - ϵ_0

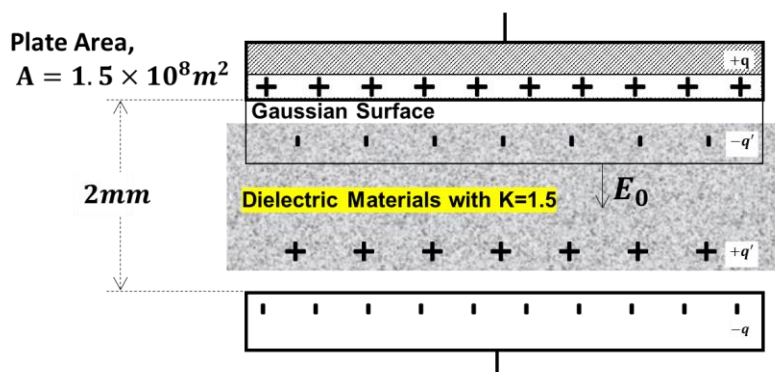
6. Which one is the unitless quantity?
- Lorentz force
 - Drift velocity
 - Conductance
 - Relative permittivity
 - Capacitance
7. Which one is the form of Ohm's law?
- $E = \rho J$
 - $J = \rho E$
 - $F = \rho J$
 - $J = \rho F$
 - $E = JA$
8. The resistance is the length of the conductor.
- inversely proportional to
 - inversely proportional to the square of
 - directly proportional to
 - directly proportional to the square of
 - equal to
9. $5000G = ? \text{ Tesla}$
- 50
 - 10
 - 5
 - 0.5
 - 0.05
10. What is the value of $\frac{1}{4\pi\epsilon_0}$?
- $1.6 \times 10^{-16} Nm^{-1}$
 - $8.85 \times 10^{-12} C^2 N^{-1} m^{-2}$
 - $9 \times 10^9 Nm^2 C^{-2}$
 - $6.6 \times 10^{-16} Nm^{-1}$
 - $1.6 \times 10^{-19} Nm^{-1}$

(Answer Question No. 1 and any three from the rest.)

[The figures in the right margin indicate full marks for the respective question]

1. See the stem carefully and answer the question that follows:

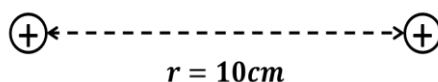
The diagram shows a parallel plate capacitor with dielectric



In Physics class, a student 'Asif' said that "Free Charge, $q < \text{Induced Charge, } q'$ "

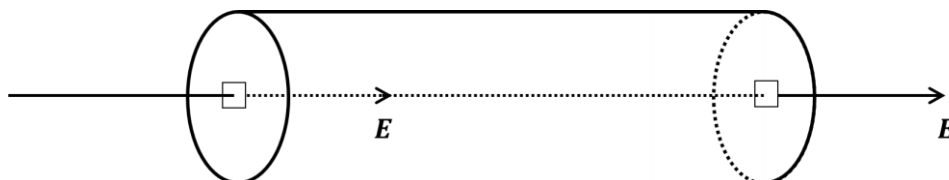
- a) Specify K. [1]
 - b) Calculate the capacitance of the mentioned capacitor. [2]
 - c) Is it true Asif's statement? If not, give your mathematical logic. [2]
- a) Clarify relative permittivity. [1]
 - b) Draw the electric lines of force for two positive charges placed in nearer. [2]
 - c) Show the figure of the arrangement of two charges – [2]

$$q_1 = 1.0 \times 10^{-6} \text{ C} \qquad q_2 = 2.0 \times 10^{-6} \text{ C}$$



At what point the line joining the above two charges the electric field strength is zero?

- a) Define 1 Farad. [1]
 - b) What is current density? Relate it with charge density. [2]
 - c) How is the resistivity related to the number of electrons per unit volume? [2]
- a) State the Gauss law. [2]
 - b) Calculate the electric flux for the following shaped object placed in an external field: [3]



- a) Formulate the vector form of Lorentz force. [1]
- b) Write the different equivalent units of magnetic field. [1]
- c) When a 10cm long wire carrying 10A current is at an angle of 30° with a magnetic field then it experiences 5N force. Find the magnitude of the magnetic field. [3]